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DEVICE AND METHOD FOR REDUCING OVERSHOOT VOLTAGE

Abstract

A device and a method of sensing and compensating an overshoot voltage are disclosed. The device includes a power converter and an overshoot voltage compensation circuit. The overshoot voltage compensation circuit includes an overshoot voltage sensing circuit to generate a first comparison voltage by performing a filtering operation on the output voltage, and an overshoot voltage control circuit to generate a first control signal for controlling the operation of a power transistor based on the first comparison voltage. The first control signal may be a signal that controls the operation of the first power transistor to change a slope of a current flowing through an inductor included in the power converter versus time.

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Background/Summary

CROSS-REFERENCE [0001] This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0030912, filed on Mar. 4, 2024 in the Korean Intellectual Property Office (KIPO), the disclosure of which is incorporated by reference herein in its entirety.

FIELD

[0002] The present disclosure generally relates to power converters, and more specifically relates to a device and method for sensing and compensating an overshoot voltage in a power converter.

DISCUSSION

[0003] Power converters may convert alternating current (AC) to AC, AC to direct current (DC), DC to DC, and/or DC to AC. DC to DC power converters may convert DC power to DC power such as by using a linear regulator, a voltage regulator, a motor-generator, a rotary converter, or a switched-mode power supply.

[0004] Due to their high power efficiency, switched-mode power supply types of power converters may be used to generate supply voltages that provide power to various electrical components or loads, and the loads may operate by consuming load currents provided by the power converters.

[0005] In order to increase the operational speed and capacity of a load, the load current may be increased and the supply voltage may be decreased for high power efficiency. When a high load current is suddenly supplied, a voltage other than a required supply voltage may occur, which may be referred to as an overshoot voltage. When an overshoot voltage occurs, the power converter might provide a relatively unstable supply voltage to the load. Thus, a device and method capable of compensating the overshoot voltage is desired.

SUMMARY

[0006] The present disclosure provides a device and a method for sensing an overshoot voltage by performing a filtering operation on an output voltage of a power converter, and controlling the sensed overshoot voltage by adjusting a slope of a current flowing through an inductor included in the power converter versus time.

[0007] In accordance with an embodiment of the present disclosure, a device comprises a power converter including an inductor, a first power transistor configured to decrease a current flowing through the inductor, a second power transistor configured to increase the current flowing through the inductor, and an output node configured to output an output voltage responsive to the current flowing through the inductor; and an overshoot voltage compensation circuit including an overshoot voltage sensing circuit configured to generate a first comparison voltage by performing a filtering operation on the output voltage, and an overshoot voltage control circuit configured to generate a first control signal for controlling operation of at least one of the first power transistor or the second power transistor based on the first comparison voltage to change a slope of the current flowing through the inductor versus time.

[0008] In accordance with an embodiment of the present disclosure, a device comprises a power converter including an inductor, a first power transistor configured to decrease a current flowing through the inductor, a second power transistor configured to increase a current flowing through the inductor, and an output node, wherein the power converter is configured to output, to the output node, an output voltage generated by stepping downing an input voltage; a high-pass filter

configured to perform a filtering operation on the output voltage to generate a filtered voltage; a first comparator configured to compare the filtered voltage with a first reference voltage to output a first comparison voltage; and an overshoot voltage control circuit configured to generate a control signal based on the first comparison voltage, wherein the control signal is a signal for controlling the power converter to perform an operation of diverting a current flowing through the inductor.

[0009] In accordance with an embodiment of the present disclosure, a method of operating a device including a power converter comprises performing a filtering operation on an output voltage of the power converter; sensing an overshoot voltage by comparing the filtered output voltage generated based on the filtering operation with a reference voltage; and compensating an overshoot voltage sensed through an operation of diverting a current flowing through an inductor included in the power converter, wherein the operation of diverting the current flowing through the inductor is an operation of changing a slope of a current flowing through the inductor versus time during a period in which the magnitude of a current applied to a load device connected to the power converter decreases.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] In order to better understand the drawings of the present disclosure as described in the detailed description, a brief description of each drawing is provided. Embodiments of the present disclosure will be more clearly understood from the following detailed description when taken in conjunction with the accompanying drawings, in which:

[0011] FIG. 1A is a block diagram schematically illustrating a device for compensating an overshoot voltage, according to an embodiment;

[0012] FIG. 1B is a graphical diagram for explaining an overshoot voltage;

[0013] FIG. 2 is a block diagram illustrating a device including an overshoot voltage compensation circuit, according to an embodiment;

[0014] FIG. 3 is a graphical diagram for describing an operation of compensating an overshoot voltage of a device including an overshoot voltage compensation circuit, according to an embodiment;

[0015] FIG. 4 is a block diagram illustrating a device including an overshoot voltage compensation circuit, according to an embodiment;

[0016] FIG. 5A is a block diagram illustrating a device including an overshoot voltage compensation circuit, according to an embodiment;

[0017] FIGS. 5B and 5C are graphical diagrams for explaining operations according to modes of a device including an overshoot voltage compensation circuit, according to an embodiment;

[0018] FIG. 6 is a block diagram illustrating a device including an overshoot voltage compensation circuit, according to an embodiment;

[0019] FIG. 7A is a block diagram illustrating a device including an overshoot voltage compensation circuit, according to an embodiment;

[0020] FIG. 7B is a graphical diagram for describing an operation of compensating an overshoot voltage of a device including an overshoot voltage compensation circuit, according to an embodiment;

[0021] FIG. 8 is a block diagram illustrating a device including an overshoot voltage compensation circuit, according to an embodiment;

[0022] FIG. 9 is a flowchart diagram illustrating a method for compensating an overshoot voltage, according to an embodiment; and

[0023] FIG. 10 is a block diagram illustrating an electronic device according to an embodiment.

DETAILED DESCRIPTION

[0024] Hereinafter, illustrative embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

[0025] As shown in FIG. 1A, a device for compensating an overshoot voltage according to an embodiment is indicated generally by the reference numeral **10**.

[0026] As shown in FIG. 1B, an overshoot voltage with respect to a plot of current versus time is indicated generally by the reference numeral **20**.

[0027] Referring to FIG. 1A, the device **10** may include a power converter **100** and an overshoot voltage compensation circuit **200**. The power converter **100** may generate an output voltage V_o based on an input voltage V_{IN} and supply the output voltage V_o generated as a supply voltage to a load through an output node **N1**. In an embodiment, the power converter **100** may be a buck-converter that lowers the input voltage $V_{sub.IN}$ to supply the lowered output voltage V_o to the load. For example, the power converter **100** may be an adaptive on time (AOT) buck-converter that has relatively fast dynamic characteristics. It shall be understood that the present disclosure is not limited to buck-converters, and may similarly be applied to boost converters, buck-boost converters, switched-mode power supply converters, and/or other types of power converters as presently known or later developed in the pertinent field of art to meet various application criteria.

[0028] The power converter **100** may include an inductor L , a first power transistor **110**, a second power transistor **120**, and an output capacitor C_o . The first power transistor **110** and the second power transistor **120** may be configured to increase or decrease via a current $I_{sub.L}$ flowing through the inductor L . For example, the first power transistor **110** may include a first metal-oxide-semiconductor field-effect transistor (MOSFET) **111** and a first diode **112**, and when a turn-on operation is performed by receiving a signal (e.g., a high level logic signal) from a gate terminal of the first MOSFET **111**, the current $I_{sub.L}$ may be decreased through the inductor L . For example, the second power transistor **120** may include a second MOSFET **121** and a second diode **122**, and when a turn-on operation is performed by receiving a signal (e.g., a high level logic signal) from a gate terminal of the second MOSFET **121**, the current $I_{sub.L}$ may be increased through the inductor L . The power converter **100** may increase or decrease the current $I_{sub.L}$ through the inductor L and supply the output voltage V_o to the load through the output node **N1**. A current source $I_{sub.load}$ may denote the load. In an embodiment, the diodes **112** and **122** may be body diodes or the like of MOSFETs **111** and **121**, respectively. In an embodiment, the first and second power transistors may include transistors **111** and **121**, which need not be bodied MOSFETs, as well as diodes **112** and **122**, each implemented separately.

[0029] In an embodiment, the load may be a memory device, without limitation thereto. For example, the load may be a dynamic RAM (DRAM). In the following specification, a DRAM load is generally provided as an example, but the present disclosure is not limited thereto. For example, the load may include volatile memory such as static random access memory (SRAM) and/or synchronous DRAM (SDRAM), and/or nonvolatile memory such as phase-change RAM (PRAM), magneto-resistive RAM (MRAM), Resistive RAM (ReRAM), and/or ferro-electric RAM (FRAM).

[0030] When the power converter **100** supplies the output voltage V_o to the load, an overshoot voltage may occur. The overshoot voltage may be a voltage greater than the output voltage V_o required by the load, and an overshoot voltage greater than or equal to an error range may cause the power converter **100** to supply a relatively unstable voltage to the load.

[0031] Referring further to FIG. 1B, the overshoot voltage **20** is based currents $I_{sub.LOAD}$ and $I_{sub.L}$, where the horizontal axis may represent time t , and the vertical axis may represent the magnitude of current I . A first period **T1** may be a period in which the power converter **100** supplies a constant load current $I_{sub.load}$, and a second period **T2** may be a period in which the load current $I_{sub.load}$ is reduced. In the second period **T2**, a difference may occur between a speed at which the current $I_{sub.L}$ of the inductor L is decreased and a speed at which the load current $I_{sub.load}$ is decreased and, accordingly, a difference in an amount of charge corresponding to an area of a first region "a" may accumulate in an output capacitor. The amount of increase may be

calculated according to the following equation.

[00001] $Q = C_O \cdot \text{Math. } V_{OS}$ [Equation1]

[0032] ΔQ may be the difference or delta in an amount of charge, C_O may be a capacity of an output capacitor C_O , and V_{OS} may be an overshoot voltage. Here, as the load current $I_{\text{sub.load}}$ drawn by the load increases, the delta increase amount ΔQ may increase. The load and the device **10** may be implemented as one or more integrated circuit chips, and may be mounted on a printed circuit board. In connection with an increase in the operational speed and capacity of the load, the size of the load may be increased, but the size of the printed circuit board may be limited.

Accordingly, the size of the device **10** may be limited, and the size of the output capacitor C_O need not be increased in order to increase the capacity of the output capacitor C_O depending on the limited size. Since the capacity of the output capacitor C_O is not increased, the overshoot voltage V_{OS} may increase as the amount of delta increase (ΔQ) is increased. Since the increased overshoot voltage V_{OS} may degrade the performance of the power converter **100**, the embodiment of FIG. 1A may be applied without increasing the size of the output capacitor C_O , and the overshoot voltage compensation circuit **200** may sense an overshoot voltage V_{OS} and may reduce the sensed overshoot voltage V_{OS} .

[0033] In greater detail, the overshoot voltage compensation circuit **200** may include an overshoot voltage sensing circuit **210** and an overshoot voltage control circuit **220**.

[0034] The overshoot voltage sensing circuit **210** may be connected to the output node **N1** of the power converter **100** to receive the output voltage V_O , and may generate a first comparison voltage by performing a filtering operation on the output voltage V_O .

[0035] In an embodiment, the overshoot voltage sensing circuit **210** may include a high pass filter and pass only an alternating current component of the output voltage V_O . The overshoot voltage sensing circuit **210** may compare an alternating current (AC) component of the output voltage V_O with a first reference voltage to generate a first comparison voltage. The first reference voltage may be a preset voltage, and may be a voltage within an error range corresponding to an alternating current component of the output voltage V_O . The output voltage V_O may include a ripple voltage for stability, and the magnitude of the direct current (DC) component of the output voltage V_O may be changed by the ripple voltage. Accordingly, when the output voltage V_O is compared with the reference voltage, an error range might be relatively large, and an overshoot voltage might be sensed relatively imprecisely. But since the overshoot voltage sensing circuit **210** generates a first comparison voltage by comparing the alternating current component of the output voltage V_O with the first reference voltage, an error range may be reduced and the overshoot voltage may be sensed relatively precisely. In other words, the overshoot voltage sensing circuit **210** may sense even a relatively low overshoot voltage.

[0036] The overshoot voltage control circuit **220** may receive the first comparison voltage from the overshoot voltage sensing circuit **210** and generate a first control signal **CS1** for controlling the operation of the first power transistor **110** based on the first comparison voltage.

[0037] In an embodiment, the first control signal **CS1** may be a signal that controls the operation of the first power transistor **110** to change the slope of the current $I_{\text{sub.L}}$ flowing through the inductor L . For example, the power converter **100** may receive a first control signal **CS1** from the overshoot voltage control circuit **220** and apply a signal (e.g., a low level logic signal) to the gate terminal of the first MOSFET **111** based on the first control signal **CS1**. The first MOSFET **111** may be turned off according to a signal applied to a gate terminal, and a current I_L may flow through the first diode **112**. The current $I_{\text{sub.L}}$ may be diverted by the first diode **112**, and a slope of the current $I_{\text{sub.L}}$ flowing through the inductor L may increase in a negative direction or become downwardly steeper in the second period **T2**. Accordingly, since the amount of delta increase (ΔQ) may be reduced, the overshoot voltage V_{OS} may be reduced.

[0038] Since the overshoot voltage sensing circuit **210** of the overshoot voltage compensation circuit **200** according to an embodiment senses an overshoot voltage based on the alternating

current component of the output voltage V_o by performing a filtering operation, a relatively low overshoot voltage may be sensed, and since the overshoot voltage compensation circuit **200** according to an embodiment generates a first control signal **CS1** for controlling the first MOSFET **111** to turn off, a current flowing through the inductor through the first diode **112** may be diverted. Accordingly, the overshoot voltage may be reduced without increasing the size of the output capacitor C_o . Accordingly, the device **10** may provide a relatively stable supply voltage to a load. [0039] Turning to FIG. 2, a device including an overshoot voltage compensation circuit according to an embodiment is indicated generally by the reference numeral **10a**. In an embodiment, a device **10a** of FIG. 2 may be an example of the device **10** of FIG. 1A.

[0040] Referring to FIGS. 1A and 2, a device **10a** may include a power converter **100a** and an overshoot voltage compensation circuit **200a**. The power converter **100a** may include an inductor L , an output capacitor C_o , a first power transistor **110a**, a second power transistor **120a**, a ripple injection circuit **130a**, a controller **140a**, and a driver **150a**. The inductor L , the output capacitor C_o , the first power transistor **110a**, and the second power transistor **120a** may be substantially the same as the inductor L , the output capacitor C_o , the first power transistor **110**, and the second power transistor **120** of FIG. 1A. Substantially redundant description to the description provided with reference to FIG. 1A may be omitted.

[0041] The ripple injection circuit **130a** may be a circuit for applying a ripple voltage to the output voltage V_o , such as for stability of the output voltage V_o , and the controller **140a** may receive a signal from the ripple injection circuit **130a** and generate a signal for controlling the driver **150a**. The driver **150a** may apply an operational signal **S1** or **S2** to the first power transistor **110a** or the second power transistor **120a** based on a signal received from the controller **140a** and/or a signal received from the overshoot voltage compensation circuit **200a**, such as, for example, based on a signal from a logic gate of the overshoot voltage compensation circuit **200a**, without limitation thereto.

[0042] The overshoot voltage compensation circuit **200a** may include an overshoot voltage sensing circuit **210a** and an AND gate **220a**. The overshoot voltage sensing circuit **210a** may include a high pass filter **211a** and a first comparator **212a**. In an embodiment, the high pass filter **211a** may include a capacitor $C_{sub.HPF}$ and a resistor $R_{sub.HPF}$, and may receive the output voltage V_o as an input and perform a high pass filtering operation on the output voltage V_o to generate a filtered voltage $\Delta V_{sub.o}$. The filtered voltage $\Delta V_{sub.o}$ may be a voltage including an alternating current component of the output voltage V_o . Although an overshoot voltage compensation circuit embodiment including at least one AND gate is shown and described, embodiments are not limited thereto. For example, in an alternate embodiment, the overshoot voltage compensation circuit may include at least one NOR gate.

[0043] In an embodiment, the first comparator **212a** may receive the filtered voltage $\Delta V_{sub.o}$ and the first reference voltage $V_{sub.REF1}$ as inputs and compare the same with each other to generate a first comparison voltage V_c . For example, when the filtered voltage $\Delta V_{sub.o}$ is greater than the first reference voltage $V_{sub.REF1}$, the first comparator **212a** may output the first comparison voltage V_c as a low level logic signal (e.g., 0 V voltage) during a period **T3** in the signal plot **30c** of FIG. 3, infra. For example, when the filtered voltage $\Delta V_{sub.o}$ is less than the first reference voltage $V_{sub.REF1}$, the first comparator **212a** may output the first comparison voltage V_c as a high level logic signal (e.g., a voltage exceeding 0 V) during a period before **T3** overlapping the period **T1** and a beginning portion of the period **T2** in the signal plots **30c** and **30d** of FIG. 3, infra.

[0044] In an embodiment, the AND gate **220a** may be an example of the overshoot voltage control circuit **220** of FIG. 1A. The controller **140a** may generate a signal **ON1** as an input to the AND gate **220a** which outputs a first control signal **CS1** as an input to control the driver **150a** to output a signal **S1** for turning on or off the first power transistor **110a**. The AND gate **220a** may receive the signal **ON1** from the controller **140a** and a first comparison voltage V_c from the first comparator **212a**, and output a first control signal **CS1** based on the result. For example, when both the signal

ON1 and the first comparison voltage V_c are a high level logic signal (e.g., voltage exceeding 0 V), the first control signal CS1 of the high level logic signal (e.g., voltage exceeding 0 V) may be output. For example, when at least one of the signal ON1 or the first comparison voltage V_c is a low level logic signal (e.g., 0 V voltage), the first control signal CS1 of a low level logic signal (e.g., 0 V voltage) may be output.

[0045] In an embodiment, the driver 150a may apply an operational signal S1 or S2 to the first power transistor 110a or the second power transistor 120a based on a signal ON2 received from the controller 140a or the first control signal CS1 received from the AND gate 220a. The controller 140a may generate a signal ON2 which is applied to the driver 150a which outputs a signal S2 for turning on or off the second power transistor 120a.

[0046] For example, when receiving a signal ON2 of a high level logic signal (e.g., a voltage exceeding 0 V) from the controller 140a, the driver 150a may generate a signal S2 of a high level logic signal (e.g., a voltage exceeding 0 V) and apply the signal S2 to the gate terminal of the second MOSFET 121a, and may turn on the second power transistor 120a. For example, when receiving a signal ON2 of a low level logic signal (e.g., a voltage of 0 V) from the controller 140a, the driver 150a may generate a signal S2 of a low level logic signal (e.g., a voltage of 0 V) and apply the signal S2 to the gate terminal of the second MOSFET 121a, and may turn off the second power transistor 120a.

[0047] For example, when receiving the first control signal CS1 of a high level logic signal (e.g., a voltage exceeding 0 V) from the AND gate 220a, the driver 150a may generate a signal S1 of a high level logic signal (e.g., a voltage exceeding 0 V) and apply the signal S1 to the gate terminal of the first MOSFET 111a, and may turn on the first power transistor 110a.

[0048] For example, when receiving the first control signal CS1 of a low level logic signal (e.g., a voltage of 0 V) from the AND gate 220a, the driver 150a may generate the signal S1 of a low level logic signal (e.g., a voltage of 0 V) and apply the signal S1 to the gate terminal of the first MOSFET 111, and may turn off the first power transistor 110a.

[0049] When an overshoot voltage is sensed, the overshoot voltage compensation circuit 200a may control the first power transistor 110a to be turned off by the controller 140a, which does not transmit, to the driver 150a, a signal ON1 generated to turn on the first power transistor 110a and transmit the first control signal CS1 generated to turn off the first power transistor 110a to the driver 150a through the AND gate 220a. Accordingly, since the device 10a may divert the current $I_{sub.L}$ flowing through the inductor L, the overshoot voltage may be reduced without increasing the size of the output capacitor C_o . In other words, the device 10a may stably provide a supply voltage to a load.

[0050] Turning now to FIG. 3, an operation of compensating an overshoot voltage of a device including an overshoot voltage compensation circuit according to an embodiment is indicated generally by the reference numeral 30.

[0051] Referring to FIGS. 2 and 3, a first signal plot 30a shows an output voltage V_o versus time, a second signal plot 30b shows a filtered voltage $\Delta V_{sub.o}$ versus time, a third signal plot 30c shows a first comparison voltage V_c versus time, and a fourth signal plot 30d shows a reduced overshoot voltage versus time.

[0052] A first period T1 and a second period T2 of the signal plot 30d may be substantially the same as the first period T1 and the second period T2 of FIG. 1B, and redundant descriptions are omitted. In the first period T1, the output voltage V_o is constant and the filtered voltage $\Delta V_{sub.o}$ is also constant with a value less than the first reference voltage $V_{sub.REF1}$, so the first comparison voltage V_c may be a high level logic signal (e.g., a voltage exceeding 0 V).

[0053] In the second period T2, the output voltage V_o increases and the filtered voltage $\Delta V_{sub.o}$ also increases, so that the filtered voltage may have a value greater than the first reference voltage $V_{sub.REF1}$, and thus, the first comparison voltage V_c may be a low level logic signal (e.g., 0 V voltage) while the filtered voltage $\Delta V_{sub.o}$ has a value greater than the first reference voltage

V.sub.REF1.

[0054] In an embodiment, when an overshoot voltage is sensed (e.g., in the second period T2), the overshoot voltage compensation circuit **200a** may control the first power transistor **110a** to be turned off by transmitting the first control signal CS1 generated to turn off the first power transistor **110a** to the driver **150a** through the AND gate **220a**, without transmitting, to the driver **150a**, a signal ON1 generated to turn on the first power transistor **110a**. When the first power transistor **110a** is turned off, the device **10a** may divert the current IL flowing through the inductor L because a current flows through the first diode **112a**.

[0055] For example, in the second period T2, the slope of the current I.sub.L flowing through the inductor L may have a greater value in a negative direction than the slope of the current I.sub.L flowing through the inductor L when the first power transistor **110a** is turned on, and may have a smaller area of the second region “b” than the area of the first region “a” of FIG. 1B. In other words, the amount of increase (ΔQ) corresponding to the area of the third region “c” may be reduced based on Equation 1 described with reference to FIG. 1B. Accordingly, the overshoot voltage may be reduced without increasing the size of the output capacitor Co, so that the device **10a** may stably provide a supply voltage to the load.

[0056] As shown in FIG. 4, a device including an overshoot voltage compensation circuit according to an embodiment is indicated generally by the reference numeral **10b**. In an embodiment, a device **10b** of FIG. 4 may be an example of the device **10a** of FIG. 2.

[0057] Referring to FIGS. 2 and 4, a device **10b** may include a power converter **100b** and an overshoot voltage compensation circuit **200b**. The power converter **100b** may include an inductor L, an output capacitor Co, a first power transistor **110b**, a second power transistor **120b**, a ripple injection circuit **130b**, a controller **140b**, and a driver **150b**. The inductor L, the output capacitor Co, the first power transistor **110b**, the second power transistor **120b**, and the ripple injection circuit **130b** of FIG. 4 may be substantially the same as the inductor L, the output capacitor Co, the first power transistor **110a**, the second power transistor **120a**, and the ripple injection circuit **130a** of FIG. 2, and a redundant description with that provided with reference to FIG. 2 may be omitted.

[0058] The controller **140b** may generate a signal ON1 or ON2 for controlling the first power transistor **110b** or the second power transistor **120b** to be turned on or off.

[0059] The overshoot voltage compensation circuit **200b** may include an overshoot voltage sensing circuit **210b**, a first AND gate **221b**, and a second AND gate **222b**, and the overshoot voltage sensing circuit **210b** may include a high pass filter **211b** and a first comparator **212b**. The high pass filter **211b** and the first comparator **212b** of FIG. 4 may be substantially the same as the high pass filter **211a** and the first comparator **212a** of FIG. 2, and a redundant description with that provided with reference to FIG. 2 may be omitted.

[0060] In an embodiment, the first AND gate **221b** and the second AND gate **222b** of FIG. 4 may be examples of the overshoot voltage control circuit **220** of FIG. 1A. The first AND gate **221b** of FIG. 4 may be substantially the same as the AND gate **220a** of FIG. 2, and a redundant description with that provided with reference to FIG. 2 may be omitted.

[0061] In an embodiment, the second AND gate **222b** may receive a signal ON2 from the controller **140b** and a first comparison voltage Vc from the first comparator **212a**, and output a second control signal CS2 based on the same. For example, when both the signal ON2 and the first comparison voltage Vc are a high level logic signal (e.g., voltage exceeding 0 V), the second control signal CS2 of the high level logic signal (e.g., voltage exceeding 0 V) may be output. For example, when at least one of the signal ON2 or the first comparison voltage Vc is a low level logic signal (e.g., 0 V voltage), the second control signal CS2 of a low level logic signal (e.g., 0 V voltage) may be output.

[0062] In an embodiment, the driver **150b** may receive the control signals (e.g., CS1 and CS2) from the first AND gate **221b** and the second AND gate **222b**, and based on this, an operational signal S1 or S2 may be applied to the first power transistor **110b** or the second power transistor

120b. The operation of the driver **150c** according to the first control signal may be substantially the same as the operation of the driver **150a** described with reference to FIG. 2, and redundant descriptions with those provided with reference to FIG. 2 are omitted.

[0063] For example, when receiving the second control signal CS2 of a high level logic signal (e.g., a voltage exceeding 0 V) from the second AND gate **220b**, the driver **150b** may generate a signal S2 of a high level logic signal (e.g., a voltage exceeding 0 V) and apply the signal S2 to the gate terminal of the second MOSFET **121b**, and may turn on the second power transistor **120b**.

[0064] For example, when receiving the second control signal CS2 of a low level logic signal (e.g., a voltage of 0 V) from the second AND gate **222b**, the driver **150b** may generate the signal S2 of a low level logic signal (e.g., a voltage of 0 V) and apply the signal S2 to the gate terminal of the second MOSFET **121b**, and may turn off the second power transistor **120b**.

[0065] When an overshoot voltage is sensed, the overshoot voltage compensation circuit **200b** may control the first power transistor **110b** to be turned off by the controller **140b**, which does not transmit, to the driver **150b**, a signal ON2 generated to turn on the second power transistor **120b** and transmit the second control signal CS2 generated to turn off the second power transistor **120b** to the driver **150b** through the second AND gate **222b**. Accordingly, the current IL of the inductor L may be prevented from being increased due to the input voltage V.sub.IN.

[0066] Turning to FIG. 5A, a device including an overshoot voltage compensation circuit according to an embodiment is indicated generally by the reference numeral **10c**. Turning to FIGS. 5B and 5C, operations according to modes of a device including an overshoot voltage compensation circuit are indicated generally by the reference numerals **50b** and **50c**, respectively. In an embodiment, a device **10c** of FIG. 5A may be an example of the device **10b** of FIG. 4.

[0067] Referring to FIGS. 4 and 5A, a device **10c** may include a power converter **100c** and an overshoot voltage compensation circuit **200c**. The power converter **100c** may be substantially the same as the power converter **100b** of FIG. 4, and a redundant description with that provided with reference to FIG. 4 may be omitted. The overshoot voltage compensation circuit **200c** of FIG. 5A may be a circuit in which a mode determination circuit **230c** is added to the overshoot voltage compensation circuit **200b** of FIG. 4, and a redundant description with that provided with reference to FIG. 4 may be omitted.

[0068] In an embodiment, the power converter **100c** may operate in either a discontinuous current mode (DCM) or a continuous current mode (CCM). For example, the first power transistor **110c** and the second power transistor **120c** may be alternately turned on or off, and in this case, when the load current I.sub.load is greater than half of the change amount of the current I.sub.L flowing through the inductor L of the power converter **100c** compared with the load current I.sub.load, the power converter **100c** may operate in a DCM, and when the load current I.sub.load is less than half of the change amount of the current I.sub.L, the power converter **100c** may operate in a CCM.

[0069] The mode determination circuit **230c** may determine whether the power converter **100c** operates in either a DCM or a CCM. Configuration and operation of the mode determination circuit **230c** may be described in greater detail further below with reference to FIG. 6.

[0070] In an embodiment, the overshoot voltage compensation circuit **200c** may perform an overshoot voltage reduction operation according to a mode determined by the mode determination circuit **230c**. The overshoot voltage reduction operation may be an operation of generating a signal for controlling the operation of the first power transistor **110c** to change the slope of the current I.sub.L flowing through the inductor L.

[0071] For example, the overshoot voltage compensation circuit **200c** may perform an overshoot voltage reduction operation on the sensed overshoot voltage when the mode determination circuit **230c** determines the mode of the power converter **100c** as a CCM.

[0072] For example, the overshoot voltage compensation circuit **200c** may not perform an overshoot voltage reduction operation on the sensed overshoot voltage when the mode determination circuit **230c** determines the mode of the power converter **100c** as a DCM.

[0073] Referring further to FIGS. 5B and 5C, a signal plot 50b and a signal plot 50c show the output voltage V_o and the filtered voltage $\Delta V_{sub.o}$ versus time when the power converter 100c operates in a DCM. The vertical axes of the signal plot 50b and the signal plot 50c may represent the magnitude of a voltage, and the horizontal axes may represent time t . When the power converter 100c operates in a DCM, the frequency of the output voltage V_o may decrease as the load current $I_{sub.load}$ decreases, and accordingly, distortion may occur in the filtered voltage $\Delta V_{sub.o}$. Distortion of the filtered voltage $\Delta V_{sub.o}$ may be prevented by adjusting the capacitor $C_{sub.HPF}$ and the resistor $R_{sub.HPF}$ included in the overshoot voltage sensing circuit 210c, but when the load current $I_{sub.load}$ is less than or equal to a threshold current, distortion may occur in the filtered voltage $\Delta V_{sub.o}$ even when the capacitor $C_{sub.HPF}$ and the resistor $R_{sub.HPF}$ included in the overshoot voltage sensing circuit 210c are adjusted. The threshold current may refer to a current at which distortion of the filtered voltage $\Delta V_{sub.o}$ occurs.

[0074] For example, the load current $I_{sub.load}$ in the signal plot 50b may be greater than a threshold current, and distortion of the filtered voltage $\Delta V_{sub.o}$ may be prevented by adjusting the capacitor $C_{sub.HPF}$ and the resistor $R_{sub.HPF}$ included in the overshoot voltage sensing circuit 210c in the signal plot 50b. The filtered voltage $\Delta V_{sub.o}$ may have a constant amplitude based on 0 V, and the overshoot voltage compensation circuit 200c may sense the overshoot voltage based on a constant reference voltage $V_{sub.REF.0V}$.

[0075] For example, the load current $I_{sub.load}$ in the signal plot 50c may be less than or equal to the threshold current, and in the section t_1 , distortion occurs in the voltage $\Delta V_{sub.o}$ filtered to the low-frequency region and may not have a constant amplitude based on 0 V. Accordingly, the overshoot voltage compensation circuit 200c may sense an overshoot voltage based on the relatively high reference voltage $V_{sub.REF.0v}$, and when the overshoot voltage is sensed based on the high reference voltage $V_{sub.REF.0v}$, the overshoot voltage improvement effect may be relatively reduced.

[0076] The overshoot voltage compensation circuit 200c may not perform an overshoot voltage reduction operation when the power converter 100c operates in a DCM, and may perform an overshoot voltage reduction operation when the power converter 100c operates in a CCM, thereby improving the overshoot voltage improvement effect.

[0077] Turning now to FIG. 6, a device including an overshoot voltage compensation circuit according to an embodiment is indicated generally by the reference numeral 10d. In an embodiment, a device 10d of FIG. 6 may be an example of the device 10c of FIG. 5A. Substantially redundant description to that provided with reference to FIG. 5A may be omitted.

[0078] Referring to FIG. 6, a mode determination circuit 230d may include a flip-flop 231d, a third AND gate 232d, an inverter 233d, and an OR gate 234d. In an embodiment, the flip-flop 231d may receive a first signal ZCS_OUT as an input, and may receive, as a clock signal CLK, a signal ON2 from a controller 140d. The first signal ZCS_OUT may have a high level logic signal (e.g., a voltage exceeding 0 V) when the magnitude of the current is less than or equal to 0, and may have a low level logic signal (e.g., a voltage of 0 V) when the magnitude of the current exceeds 0.

[0079] For example, the first signal ZCS_OUT may be generated on the basis of the current $I_{sub.L}$ flowing through the inductor L, and the flip-flop 231d may generate an output signal of a low level logic signal (e.g., 0 V voltage) when the power converter 100d operates in a DCM, and may generate an output signal of a high level logic signal (e.g., a voltage exceeding 0 V) when the power converter 100d operates in a CCM.

[0080] In an embodiment, the third AND gate 232d may receive, as inputs, a second signal ZCS_CAL_END, a third signal SS_DONE, and the output signal of the flip-flop 231d. The second signal ZCS_CAL_END and the third signal SS_DONE may be signals indicating a time point at which the power converter 100d normally operates. For example, the second signal ZCS_CAL_END may be a high level logic signal (e.g., a voltage exceeding 0 V) when the calibration for a circuit (not shown) generating the first signal ZCS_OUT is completed, and the

third signal SS_DONE may be a high level logic signal (e.g., a voltage exceeding 0 V) when the power of the device **10d** is turned on.

[0081] In an embodiment, the inverter **233d** may invert the output of the third AND gate **232d**. In an embodiment, the OR gate **234d** may receive the output of the inverter **233d** and the first comparison voltage V_c as inputs, and the first AND gate **221d** and the second AND gate **222d** may receive the output of the OR gate **234d** as an input.

[0082] For example, when the power converter **100d** operates in a DCM, the output of the third AND gate **232d** is a low level logic signal (e.g., 0 V), and thus, the OR gate **234d** may generate an output of a high level logic signal (e.g., a voltage exceeding 0 V) regardless of the first comparison voltage V_c . Accordingly, since the signal ON1 or ON2 generated by the controller **140d** may be transmitted to the driver **150d**, the overshoot voltage compensation circuit **200d** may not perform the overshoot voltage reduction operation.

[0083] For example, when the power converter **100d** operates in a CCM, the output of the third AND gate **232d** is a high level logic signal (e.g., a voltage exceeding 0 V), and thus, the OR gate **234d** may generate an output of a high level logic signal (e.g., a voltage exceeding 0 V) or a low level logic signal (e.g., a voltage of 0 V) according to the first comparison voltage V_c . In other words, the overshoot voltage compensation circuit **200d** may perform an overshoot voltage reduction operation on the sensed overshoot voltage.

[0084] The overshoot voltage compensation circuit **200d** may not perform an overshoot voltage reduction operation when the power converter **100d** operates in a DCM, and may perform an overshoot voltage reduction operation when the power converter **100d** operates in a CCM, thereby improving the overshoot voltage improvement effect.

[0085] As shown in FIG. 7A, a device including an overshoot voltage compensation circuit according to an embodiment is indicated generally by the reference numeral **10e**.

[0086] As shown in FIG. 7B, a signal plot for an operation of compensating an overshoot voltage of a device including an overshoot voltage compensation circuit according to an embodiment is indicated generally by the reference numeral **40**.

[0087] In an embodiment, a device **10e** of FIG. 7A may be an example of the device **10a** of FIG. 2. Substantially redundant description to that provided with reference to FIGS. 2 and/or 3 may be omitted.

[0088] Referring to FIGS. 2 and 7A, a device **10e** may include a power converter **100e** and an overshoot voltage compensation circuit **200e**. The power converter **100e** may be substantially the same as the power converter **100a** of FIG. 2, and substantially redundant description to that provided with reference to FIG. 2 may be omitted. The overshoot voltage compensation circuit **200e** of FIG. 7A may be a circuit in which a current diversion circuit **240e** is added to the overshoot voltage compensation circuit **200a** of FIG. 2, and as described with reference to FIG. 2, the overshoot voltage may be sensed and the sensed overshoot voltage may be reduced. For example, the overshoot voltage sensing circuit **210e** may sense an overshoot voltage relatively precisely by comparing the alternating current component of the output voltage V_o with the first reference voltage $V_{sub.REF1}$, to thereby generate a first comparison voltage, and when the overshoot voltage sensing circuit **210e** senses the overshoot voltage, the overshoot voltage control circuit **220e** may generate a first control signal CS1 that controls the first MOSFET **111e** to be turned off. Hereinafter, substantially redundant descriptions to those provided with reference to FIG. 2 are omitted.

[0089] In an embodiment, the current diversion circuit **240e** may perform an operation of diverting the current $I_{sub.L}$ flowing through the inductor L based on the filtered voltage $\Delta V_{sub.o}$. For example, the current diversion circuit **240e** may compare the filtered voltage $\Delta V_{sub.o}$ with a second reference voltage to generate a second comparison voltage, and may perform an operation of diverting the current $I_{sub.L}$ flowing through the inductor L based on the second comparison voltage. The specific structure and operation of the current diversion circuit **240e** may be described

in greater detail further below with reference to FIG. 8.

[0090] Referring further to FIG. 7B, the horizontal axis of the signal plot may represent time t , and the vertical axis may represent a magnitude of current I . A first period T1 and a second period T2 of FIG. 7B may be substantially the same as the first period T1 and the second period T2 of FIG. 1B, and redundant descriptions are omitted. When sensing an overshoot voltage (e.g., in the second period T2), the current diversion circuit 240e may perform an operation of diverting the current $I_{\text{sub.L}}$ flowing through the inductor L , and accordingly, the amount of delta increase (ΔQ) may also be reduced, so that the amount of increase corresponding to a fourth region “d” may be reduced from the first region “a” of FIG. 1B. In other words, the current diversion circuit 240e may reduce an overshoot voltage sensed by diverting the current $I_{\text{sub.L}}$ flowing through the inductor L without increasing the size of the output capacitor C_o included in the power converter 100e.

[0091] Turning to FIG. 8, a device including an overshoot voltage compensation circuit according to an embodiment is indicated generally by the reference numeral 10f. In an embodiment, a device 10f of FIG. 8 may be an example of the device 10e of FIG. 7A. Substantially redundant description to that provided with reference to FIG. 7A may be omitted.

[0092] Referring to FIG. 8, a current diversion circuit 240f may be an example of the current diversion circuit 240e of FIG. 7A, and the current diversion circuit 240f may include a second comparator 241f, a first transistor 242f, a resistor $R_{\text{sub.1}}$, a second transistor 243f, and a current source $I_{\text{sub.1}}$. Although the current diversion circuit 240f includes all of the second comparator 241f, the first transistor 242f, the resistor $R_{\text{sub.1}}$, the second transistor 243f, and the current source $I_{\text{sub.1}}$, embodiments are not limited thereto. For example, the current diversion circuit 240f may include the second comparator 241f, the first transistor 242f, and the resistor $R_{\text{sub.1}}$, or may include the second comparator 241f, the second transistor 243f, and a current source $I_{\text{sub.1}}$.

[0093] In an embodiment, the second comparator 241f may compare the filtered voltage $\Delta V_{\text{sub.o}}$ with the second reference voltage $V_{\text{sub.REF2}}$ to generate a second comparison voltage V_{c2} . The second reference voltage $V_{\text{sub.REF2}}$ may be a preset voltage. For example, when the filtered voltage $\Delta V_{\text{sub.o}}$ is greater than the second reference voltage $V_{\text{sub.REF2}}$, the second comparator 241f may determine that the overshoot voltage exists and may generate an output of a high level logic signal (e.g., a voltage exceeding 0 V). For example, when the filtered voltage $\Delta V_{\text{sub.o}}$ is less than the second reference voltage $V_{\text{sub.REF2}}$, the second comparator 241f may determine that the overshoot voltage does not exist and may generate an output of a low level logic signal (e.g., a voltage of 0 V).

[0094] In an embodiment, the first transistor 242f or the second transistor 243f may receive the output of the second comparator 241f through a gate terminal, and may perform a turn-on operation or a turn-off operation based on the same.

[0095] For example, when the first transistor 242f receives a signal of a high level logic signal (e.g., a voltage exceeding 0 V) at the gate terminal, an overshoot voltage is sensed, and thus, a turn-on operation may be performed and the current $I_{\text{sub.L}}$ flowing through the inductor L through the resistor $R_{\text{sub.1}}$ may be diverted.

[0096] For example, when the second transistor 243f receives a signal of a high level logic signal (e.g., a voltage exceeding 0 V) at the gate terminal, an overshoot voltage is sensed, and thus, a turn-on operation may be performed and the current $I_{\text{sub.L}}$ flowing through the inductor L through the current source $I_{\text{sub.1}}$ may be diverted.

[0097] When the overshoot voltage is sensed, the overshoot voltage compensation circuit 200f may turn on the first transistor 242f or the second transistor 243f to divert the current $I_{\text{sub.L}}$ flowing through the resistor $R_{\text{sub.1}}$ or the current source $I_{\text{sub.1}}$, thereby reducing the overshoot voltage without increasing the size of the output capacitor C_o .

[0098] Turning now to FIG. 9, a method for compensating an overshoot voltage according to an embodiment is indicated generally by the reference numeral 900. Referring to FIG. 9, a method 900 of compensating an overshoot voltage may include a plurality of operations S910 to S930.

[0099] Referring to FIGS. 1A and 9, in operation S910, the overshoot voltage sensing circuit **210** may perform a filtering operation on an output voltage V_o . In an embodiment, the overshoot voltage sensing circuit **210** may include a high pass filter and may perform a filtering operation of passing only an alternating current component of the output voltage V_o . The output voltage V_o may include a ripple voltage for stability, and the magnitude of the direct current component of the output voltage V_o may be changed by the ripple voltage. Accordingly, when the output voltage V_o is compared with the reference voltage, an error range may be relatively large, and an overshoot voltage may not be sensed relatively precisely. In the case of the filtered voltage, the direct current component may be removed to have a relatively less value compared to the output voltage V_o , and thus the error range may be relatively narrow.

[0100] In operation S920, the overshoot voltage sensing circuit **210** may sense the overshoot voltage based on the filtered output voltage. In an embodiment, the overshoot voltage sensing circuit **210** may compare a filtered output voltage with a first reference voltage $V_{sub.REF1}$ to generate a first comparison voltage. The first reference voltage $V_{sub.REF1}$ may be a preset voltage, and may be a voltage within an error range corresponding to an alternating current component of the output voltage V_o . Since the filtered output voltage may have a relatively less value compared to the output voltage V_o , the overshoot voltage may be sensed based on the relatively less first reference voltage $V_{sub.REF1}$. For example, when the filtered output voltage is greater than the first reference voltage $V_{sub.REF1}$, the overshoot voltage sensing circuit **210** may sense that the output voltage V_o includes an overshoot voltage. For example, when the filtered output voltage is less than the first reference voltage $V_{sub.REF1}$, the overshoot voltage sensing circuit **210** may sense that the output voltage V_o does not include an overshoot voltage.

[0101] In operation S930, the overshoot voltage control circuit **220** may reduce the sensed overshoot voltage by diverting a current flowing through the inductor. In an embodiment, when the overshoot voltage sensing circuit **210** senses the overshoot voltage, the overshoot voltage control circuit **220** may generate a first control signal CS1 for controlling the operation of the first power transistor **110** to divert the current $I_{sub.L}$ flowing through the inductor L.

[0102] As shown in FIG. 10 is a block diagram illustrating an electronic device according to an embodiment.

[0103] Referring to FIG. 10, an electronic device is indicated generally by the reference numeral **1000**. The electronic device **1000** may include an application processor (AP) **1010**, a transceiver **1020**, a memory **1030**, a display **1040**, and an input/output (I/O) device **1050**.

[0104] The AP **1010** may control the overall operation of the electronic device **1000** and the operation of the components of the electronic device **1000**. The AP **1010** may perform various operations. According to an embodiment, the AP **1010** may include a single processor core or a plurality of processor cores (i.e., a multi-core).

[0105] The electronic device **1000** may communicate with the outside through the transceiver **1020**. The transceiver **1020** may be, for example, a wireless short-range communication interface, such as a wired local area network (LAN), Bluetooth, Wireless Fidelity (Wi-fi), and Zigbee, or a modem communication interface capable of accessing a mobile cellular network, such as Power Line Communication (PLC), 3rd Generation (3G), Long Term Evolution (LTE), 5G, NR, and next-generation communication.

[0106] The memory **1030** may store an instruction code, control data, or user data for controlling the electronic device **1000**. The memory **1030** may include at least one of a volatile memory and a nonvolatile memory.

[0107] The display **1040** may display internal state information of the electronic device **1000**. The display **1040** may include a touch sensor (not shown). In addition, the display **1040** may include input or output functions and external appearances for a user interface. The user may control the electronic device **1000** through the touch sensor and the user interface.

[0108] The I/O device **1050** may include input means such as a touch pad, a keypad, an input

button, and the like, and output means such as a display, a speaker, and the like.

[0109] At least some of the components of the electronic device **1000**, for example, the AP **1010**, the transceiver **1020**, the memory **1030**, the display **1040**, and the I/O device **1050** may include an overshoot voltage compensation circuit that performs a filtering operation to sense an overshoot voltage and performs an operation of reducing the sensed overshoot voltage. The overshoot voltage compensation circuit **200** according to the embodiments described above with reference to FIGS. **1A** to **9** may be applied as the circuit. In an embodiment, an overshoot voltage compensation circuit may be applied to at least one of the AP **1010**, the transceiver **1020**, the memory **1030**, the display **1040**, and/or the I/O device **1050**. In an embodiment, a plurality of overshoot voltage compensation circuits may be applied to a plurality of the AP **1010**, the transceiver **1020**, the memory **1030**, the display **1040**, and/or the I/O device **1050**.

[0110] While the inventive concept has been particularly shown and described with reference to embodiments thereof, it shall be understood by those of ordinary skill in the pertinent art that various changes in form and details may be made therein without departing from the scope and spirit of the present disclosure as set forth in the following claims.

Claims

1. A device comprising: a power converter including an inductor, a first power transistor configured to decrease a current flowing through the inductor, a second power transistor configured to increase the current flowing through the inductor, and an output node configured to output an output voltage responsive to the current flowing through the inductor; and an overshoot voltage compensation circuit including an overshoot voltage sensing circuit configured to generate a first comparison voltage by performing a filtering operation on the output voltage, and an overshoot voltage control circuit configured to generate a first control signal for controlling operation of at least one of the first power transistor or the second power transistor based on the first comparison voltage to change a slope of the current flowing through the inductor versus time.
2. The device of claim 1, wherein the overshoot voltage compensation circuit comprises: a high-pass filter that includes a capacitor and a resistor and is configured to perform a filtering operation on the output voltage to generate a filtered voltage; and a first comparator configured to generate the first comparison voltage by comparing the filtered voltage with a first reference voltage, wherein the power converter is configured to output, to the output node, an output voltage generated by stepping down an input voltage.
3. The device of claim 1, wherein the first power transistor comprises a first metal oxide semiconductor field effect transistor (MOSFET) having a first diode connected between its source and drain terminals, wherein the overshoot voltage control circuit generates the first control signal to apply a low level logic signal to a gate terminal of the first MOSFET when the first comparison voltage is a low level logic signal.
4. The device of claim 3, wherein the power converter further comprises: a driver that applies a low level logic or a high level logic operational signal to the gate terminal of the first MOSFET; and a controller configured to generate a signal for controlling the driver, wherein the overshoot voltage control circuit comprises a logic gate configured to receive the signal from the controller and the first comparison voltage as inputs and to output the first control signal to the driver.
5. The device of claim 3, wherein the logic gate comprises an AND gate or a NOR gate, wherein the overshoot voltage control circuit generates a second control signal for controlling the operation of the second power transistor so that the second power transistor is turned off, when the first comparison voltage is a low level logic signal.
6. The device of claim 1, wherein the power converter operates in either a discrete current mode (DCM) or a continuous current mode (CCM), wherein the overshoot voltage compensation circuit further comprises a mode determination circuit configured to determine whether the power

converter operates in the DCM or the CCM, wherein the overshoot voltage control circuit generates the first control signal when the mode determination circuit determines the mode of the power converter as the CCM.

7. The device of claim 6, wherein the power converter further comprises: a driver for applying a low level logic signal or a high level logic signal to the second power transistor; and a controller configured to generate a signal for controlling the driver, wherein the mode determination circuit comprises: a flip-flop receiving the generated signal from the controller as a clock signal; an AND gate receiving the output of the flip-flop as an input; an inverter that receives the output of the AND gate as an input; and an OR gate receiving the output of the inverter and the first comparison voltage as inputs.

8. The device of claim 1, wherein the overshoot voltage compensation circuit further comprises: a second comparator generating a second comparison voltage by comparing, with a second reference voltage, the voltage on which the filtering operation on the output voltage received from the overshoot voltage sensing circuit is performed; and a current diversion circuit configured to divert a current flowing through the inductor, based on the second comparison voltage.

9. The device of claim 8, wherein the current diversion circuit comprises a first transistor and a resistor, or a second transistor and a current source, wherein the first transistor or the second transistor performs a turn-on operation based on the second comparison voltage to divert the current flowing through the inductor.

10. The device of claim 1, wherein the overshoot voltage control circuit generates the first control signal during a period in which the magnitude of the current applied to a load device connected to the output node decreases.

11. A device comprising: a power converter including an inductor, a first power transistor configured to decrease a current flowing through the inductor, a second power transistor configured to increase a current flowing through the inductor, and an output node, wherein the power converter is configured to output, to the output node, an output voltage generated by stepping downing an input voltage; a high-pass filter configured to perform a filtering operation on the output voltage to generate a filtered voltage; a first comparator configured to compare the filtered voltage with a first reference voltage to output a first comparison voltage; and an overshoot voltage control circuit configured to generate a control signal based on the first comparison voltage, wherein the control signal is a signal for controlling the power converter to perform an operation of diverting a current flowing through the inductor.

12. The device of claim 11, wherein the first power transistor comprises a MOSFET and a first diode, wherein the operation of diverting the current flowing through the inductor comprises an operation of turning off the first MOSFET by applying a low level logic signal to the gate terminal of the first MOSFET based on the control signal.

13. The device of claim 12, wherein the operation of diverting the current flowing through the inductor further comprises an operation of turning off the second power transistor based on the control signal.

14. The device of claim 11, further comprising: a mode determination circuit determining whether the power converter operates in either a DCM or a CCM, wherein the overshoot voltage control circuit generates the control signal when the mode determination circuit determines the mode of the power converter as the CCM.

15. The device of claim 11, further comprising: a second comparator configured to compare the filtered voltage with a second reference voltage to output a second comparison voltage; and a current diversion circuit configured to divert a current flowing through the inductor based on the second comparison voltage.

16. The device of claim 11, wherein the overshoot voltage control circuit generates the control signal during a period in which the magnitude of the current applied to a load device connected to the output node decreases.

17. A method of operating a device including a power converter, the method comprising: performing a filtering operation on an output voltage of the power converter; sensing an overshoot voltage by comparing the filtered output voltage generated based on the filtering operation with a reference voltage; and compensating an overshoot voltage sensed through an operation of diverting a current flowing through an inductor included in the power converter, wherein the operation of diverting the current flowing through the inductor is an operation of changing a slope of a current flowing through the inductor versus time during a period in which the magnitude of a current applied to a load device connected to the power converter decreases.

18. The method of claim 17, wherein the power converter comprises: a first power transistor configured to decrease a current flowing through the inductor; and a second power transistor configured to increase a current flowing through the inductor, wherein the first power transistor comprises a first MOSFET and a first diode, and the reducing of the sensed overshoot voltage comprises turning off the first MOSFET by applying a low level logic signal to the gate terminal of the first MOSFET, and wherein the operation of diverting the current flowing through the inductor comprises an operation of diverting a current flowing through the inductor by using the first diode.

19. The method of claim 18, wherein the reducing of the sensed overshoot voltage further comprises applying a low level logic signal to the second power transistor to turn off the second power transistor.

20. The method of claim 17, wherein the reducing of the sensed overshoot voltage comprises determining whether the power converter operates in either a DCM or a CCM, and performing the operation of diverting the current flowing through the inductor, when the operation mode of the power converter is determined to be the CCM.
